

STUDENT NAME:

TEACHER NAME: BOHL CVEK FINS KERJ MCLS PHIN ROSP SCHP

MATHEMATICAL METHODS UNIT 3

QUIZ 3B (Tech-Active)

Test theme: Polynomials, Exponential and Logarithmic Functions

Reading Time: 5 minutes

Writing time: 40 minutes

QUESTION AND ANSWER BOOK

<i>Section</i>	<i>Number of Questions</i>	<i>Number of questions to be answered</i>	<i>Number of marks</i>
A	10	10	10
B	1	1	20
			Total 30

- Students are permitted to bring into the room: pens, pencils, highlighters, erasers, sharpeners, rulers, a protractor, set-squares, aids for curve sketching, one bound reference, one Mathematica bound reference, a laptop and, if desired, one scientific calculator. Calculator memory DOES NOT need to be cleared.
- Students are NOT permitted to bring into the test: blank sheets of paper or white out liquid/tape.

Materials supplied

- Question and answer book of 10 pages.
- Working space is provided throughout the book.

Instructions

- Write your **name** in the space provided above on this page.
- All written responses must be in English.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the test room.

SECTION A – Multiple-choice questions**Instructions for Section A**

Answer **all** questions by circling the correct response.

Choose the response that is **correct** for the question.

A correct answer scores 1; an incorrect answer scores 0.

Marks will **not** be deducted for incorrect answers.

No marks will be given if more than one answer is circled for any question.

Question 1

The exact value of the x -intercept of $y = 7 \log_2(1 - 6x) + 9$ is

- A. $(0.121, 0)$
- B. $(0.098, 0)$
- C. $\left(\frac{1}{6} - \frac{1}{12 \times 2^{\frac{2}{7}}}, 0\right)$
- D. $\left(\frac{1}{6} - \frac{1}{6 \times e^{\frac{9}{7}}}, 0\right)$
- E. $(9, 0)$

Question 2

$\log_e(5ab)^2 + \log_e(ba)$ expressed as a single logarithm if possible is

- A. $\log_e(5ab)^2 + \log_e(ba)$ cannot be expressed as a single logarithm
- B. $2 \log_e(5a^2b^2)$
- C. $\log_e(25ab)$
- D. $\log_e(25a^3b^3)$
- E. $\log_e(10a^3b^3)$

Question 3

The function $f : [a, \infty) \rightarrow \mathbb{R}$ with rule $f(x) = \log_e(x^4)$ will have an inverse function if

- A. $a < 0$
- B. $a \leq -1$
- C. $a \leq 1$
- D. $a > 0$
- E. $a \geq -1$

Question 4

The function g has the rule $g(x) = \log_e(x - b)$, where b is a real constant.

The maximal domain of g is

- A. R^+
- B. $R \setminus \{b\}$
- C. R
- D. $(-b, b)$
- E. (b, ∞)

Question 5

The equation $f(x) = 0$ is to be solved using the bisection method.

If $f(3) = 4$ and $y = f(x)$ has one vertical asymptote with equation $x = -4$, which one of the following is an appropriate choice for an endpoint of the initial interval?

- A. -5 such that $f(-5) = -26$
- B. 0 such that $f(0) = 5$
- C. 1 such that $f(1) = -2$
- D. 13 such that $f(13) = 2$
- E. 3 such that $f'(3) = 0$

Question 6

The graph of $y = kx - 3$ intersects the graph of $y = -x^2 + 2x - 12$ at two distinct points for

- A. $k \in (-\infty, -8) \cup (4, \infty)$
- B. $k \in (-\infty, -4) \cup (8, \infty)$
- C. $k \in (-4, 8)$
- D. $k \in [-4, -8]$
- E. $k \in [-4, 8]$

Question 7

Which one of the following is a list of all possible rational roots of the polynomial $3x^4 + 2x^3 - 28x^2 - 18x + 9$?

- A. $\frac{1}{1}, \frac{1}{3}, \frac{3}{1}, \frac{3}{3}, \frac{9}{1}, \frac{9}{3}$
- B. $3, -3, -1, \frac{1}{3}$
- C. $\pm 1, \pm 3, \pm 9, \pm \frac{1}{3}, \pm \frac{3}{3}, \pm \frac{9}{3}$
- D. $\pm 1, \pm 3, \pm 9, \pm \frac{1}{3}$
- E. $\pm 1, \pm 3, \pm \frac{1}{3}, \pm \frac{1}{9}$

Question 8

Let the function $f : R \rightarrow R$, $f(x) = e^x + e^{-x}$.

For all $u \in R$, $f(2u)$ is equivalent to

- A. $f(u) + f(-u)$
- B. $2f(u)$
- C. $(f(u))^2 - 2$
- D. $(f(u))^2$
- E. $(f(u))^2 + 2$

Question 9

If $\log_5(2.5) = p$, then $\log_5(2)$ is equal to

- A. $p - 0.5$
- B. $p - \log_5(0.5)$
- C. $1 - p$
- D. $p - 1$
- E. $1.25p$

Question 10

The graph of a polynomial function with rule $y = f(x)$ has a local maximum at the point $(a, f(a))$.

The graph also has a local minimum at $(0, 0)$ but no other stationary points.

The graph of the function with the rule $y = -2f\left(\frac{x}{2}\right) + k$, where k is a positive real number, has

- A. a local maximum at the point with coordinates $(2a, -2f(a) + k)$
- B. a local maximum at the point with coordinates $\left(\frac{a}{2}, -2f(a) + k\right)$
- C. a local maximum at the point with coordinates $(2a, -2f(a) - k)$
- D. a local minimum at the point with coordinates $\left(\frac{a}{2}, 2f(a) + k\right)$
- E. a local minimum at the point with coordinates $(2a, -2f(a) + k)$

SECTION B

Instructions

Answer **all** questions in the spaces provided.

Unless otherwise specified an **exact** answer is required to a question.

In questions where more than one mark is available, appropriate working **must** be shown.

Ada Lovelace enjoys working from home as a computer programmer, (endless snacks, wearing pyjamas all day, what's not to love.). Her cat Astro is preventing her from working today by sleeping on her laptop, so she gets to watch her chickens in the backyard instead. Life is good.

One of Ada's chickens is a breed called a silkie and her name is Queenie. She is a friendly chicken and loves pats. Queenie has far more fur than feathers - fluffy furry legs, blue ears, tiny wings (silkie's cannot fly), and a fluffy mohawk which covers her eyes. As she can't see much through her pompom head, she bumps into things quite a lot. She often gets lost and sleeps in the middle of the yard as she can't always remember where her nesting box is. Queenie is a unique chicken.



Question 1 (20 marks)

Queenie the chicken is walking around the backyard. She is on a path that she regularly follows. Ada thinks this is very interesting behaviour and decides to map Queenie's favourite path using a quartic equation. Queenie's path is defined by the function

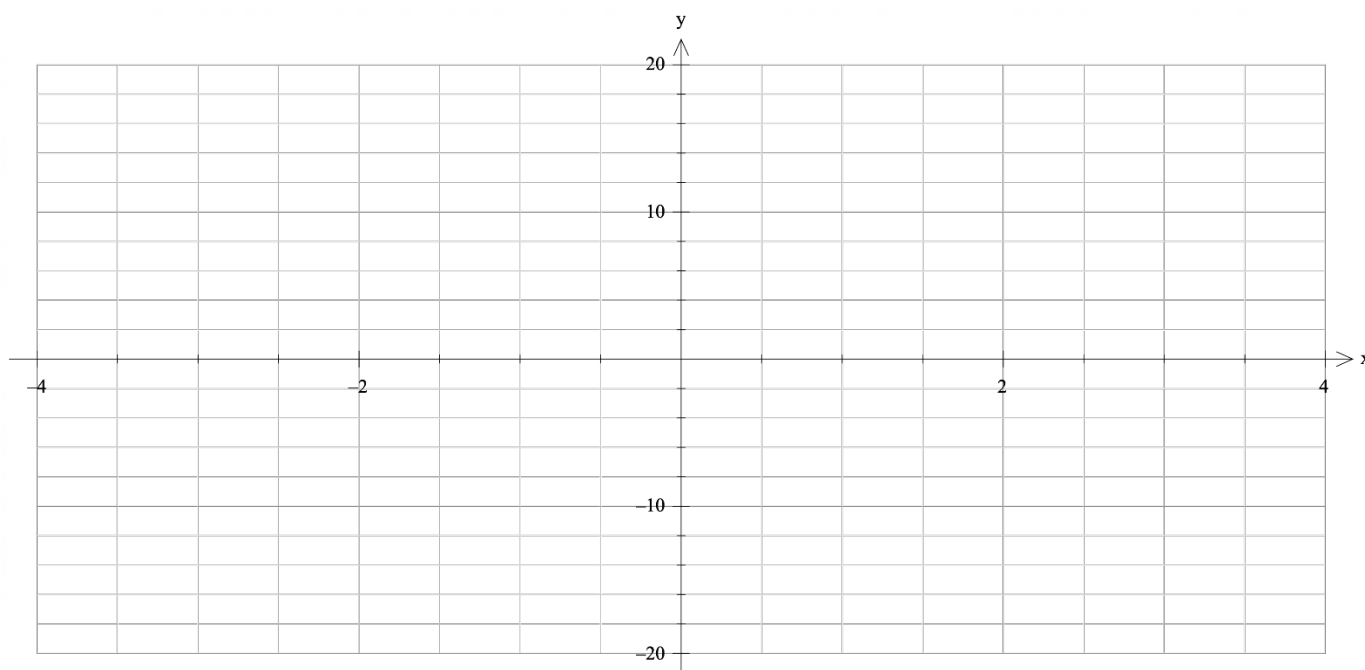
$$f : [-3, 2] \rightarrow R, \quad f(x) = \frac{3}{2}x^4 + 2x^3 - 6x^2,$$

where the x -axis represents east/west direction and the y -axis represents the north/south direction. All measurements are in metres.

- a. Sketch a graph of $y = f(x)$ on the axes below.

Label any axes intercepts, stationary points and endpoints with their coordinates.

4 marks



- b. State the values of $b \in R$ for which the graph of $y = f(x) + b$ has no x -intercepts 1 mark

Ada wants to help Queenie and decides to put some chicken seed on the x -intercepts of her favourite path.

- c. Ada decides to estimate an x -intercept by using the bisection method. She writes the following pseudocode for estimating a solution to $f(x) = 0$ using the bisection method.

Inputs: $f(x)$, a function of x , where x is a real number
 a , the lower interval endpoint
 b , the upper interval endpoint
 max , the maximum number of iterations

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Define bisection( $f(x), a, b, max$ )
  If  $f(a) \times f(b) > 0$  Then
    Return "Invalid interval"
   $i \leftarrow 1$ 
  While  $i < max$  Do
     $mid \leftarrow (a + b) \div 2$ 
    If  $f(mid) = 0$  Then
      Return  $mid$ 
    Else If  $f(a) \times f(mid) < 0$  Then
       $b \leftarrow mid$ 
    Else
       $a \leftarrow mid$ 
     $i \leftarrow i + 1$ 
  EndWhile
  Return  $mid$ 

```

The algorithm is implemented as follows:

$bisection((3/2)x^4 + 2x^3 - 6x^2, -3, -1, 4)$

- i. What is the initial interval? 1 mark

- ii. What value, correct to two decimal places, is returned when the above algorithm is implemented? 1 mark

Astro the cat has joined Queenie the chicken in the backyard. Astro also enjoys watching the chickens. Ada likes to think that Astro is being a guard-cat, however, she has her doubts.



Ada decides to map Astro's path.

- d.** Astro walks in a straight line across the yard and meets Queenie's path, $y = f(x)$, at $x = -\frac{1}{3}$.

- i.** Find $f\left(-\frac{1}{3}\right)$. 1 mark

- ii.** The straight line Astro walks in is tangent to $y = f(x)$ at $x = -\frac{1}{3}$.

Show that the equation of this line is $54y - 240x - 41 = 0$. 2 marks

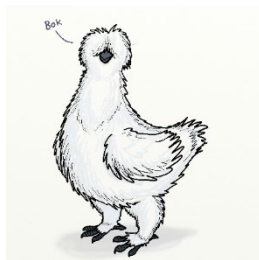
- e. i.** The straight line Astro walks in intersects $y = f(x)$ at the point where $x = -\frac{1}{3}$ and at two other points. State the x -coordinates of the two other points of intersection. Express your answers in the form $\frac{a \pm \sqrt{b}}{c}$ where a , b , and c are integers. 2 marks

- ii. Show that the distances between $x = -\frac{1}{3}$ and each intersection point found in **part e. i.** are equal, correct to two decimal places.

2 marks

Ada is not wanting to return to work and procrastinates further about Queenie's favourite path. Though, she likes to think she is practising mindfulness ... which of course she is!

Ada decides to manipulate the quartic equation with unknown parameters to change the features of the path.



Let the function $p : R \rightarrow R$, $p(x) = \frac{3}{2}x^4 + 2x^3 + 3(a-2)x^2 - 6ax + \frac{1}{2}a^2$ where a is a real constant.

- f. State the value of a for which $f(x) = p(x)$ for all x .

1 mark

- g. Find, in terms of a where appropriate, all solutions to $p'(x) = 0$.

1 mark

- h. i.** Find the values of a for which p has only one stationary point. 1 mark

- ii.** Find the minimum value of p when $a = 2$. 1 mark

- iii.** If p has only one stationary point, find the values of a for which $p(x) = 0$ has no real solutions. 2 marks

Ada has had a super productive day. She thinks Queenie the chicken has the most interesting mathematical path. Everyone should have pet chickens!



Drawn by Ada, aged 10.

A friend asked me, "How do you raise chickens?"

I answered, "Just try and wing it."